

Ocean Mixing

Scope for theory

- ❶ Where does the mixing come from?
 - Mechanical forcing: tides, wind forcing. Mesoscale eddies.
- ❷ How and why does it affect the circulation?
 - Mixing creates a diffusivity, which enables water to cross density surfaces, and to circulate.
- ❸ Having said this, there are theories of the deep circulation that *do not need eddy mixing*.
 - Role of the ACC

The quantification of all this remains an enormous theoretical and observational challenge. Without theory to guide us, modelling (and the taking of observations, I might add) is blind.